Physical Chemistry for Combustion
Chapter 1: Thermodynamics

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2017-2018





outlines

1 Introduction

2 State of matter and equations of state of gas

3 Thermodynamic systems

4 Thermodynamics laws and important quantities

5 Thermochemistry

6 Chemical equilibrium

7 Thermodynamics properties in combustion

General description

Thermodynamics

- **Describes state of a system and its most favorable evolution**
- **Predicts the final state for a given initial state**
- **Path between states is not considered**
- **Description with a number of thermodynamic properties**
- **Characterize energy content of a mixture, energy released by individual reactions, direction of reactions**

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State of matter (1)

Four states of matter

Solid

Liquid

Gas

Plasma

Differences : degree of freedom of the individual molecules, degree of organization, density, etc

For combustion : focus on gas

State of matter (2)

Gas

Large separation between the molecules

Limited impact of intermolecular forces

The molecules of a gas always have translational motion

Gas at rest : mean velocity is zero

A gas can be characterized by

- **Pressure :** force exerted by the gas molecule on a surface (in Pa)
- **Temperature :** indicates the direction of the energy flow (in K)
- **Volume :** occupied by the gas (in m^3)

Equation of state (1)

Perfect gas law

Intermolecular forces are neglected

$$PV = nRT$$

n : the number of mole

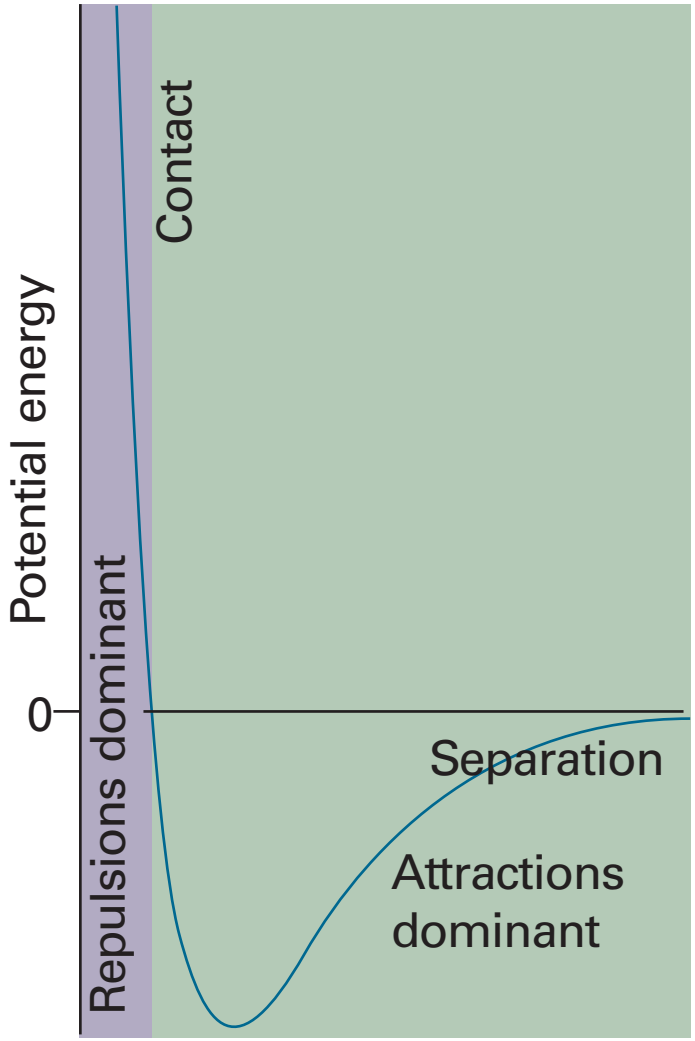
R : the perfect gad constant R=8.314 $JK^{-1}mol^{-1}$

Mixing rules

- **Dalton's law :** $P = \sum_{i=1}^{N} P_i$
- **Mole fraction :** $x_i = \frac{n_i}{\sum_{i=1}^{N} n_i}$
- **Partial pressure :** $P_i = x_i P$.

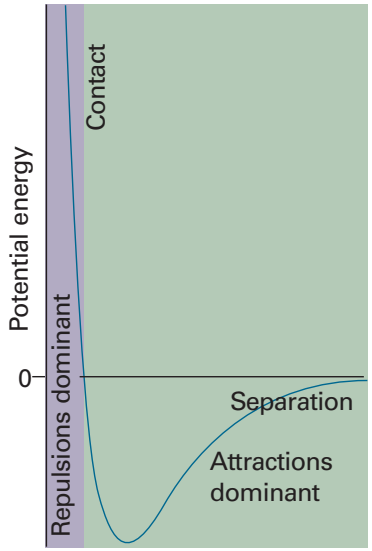
Equation of state (2)

Real gas



Equation of state (2)

Real gas



Low pressure : large distance between molecules $=>$ no interaction $=>$ perfect gas

Intermediate pressure : average distance between molecules $=>$ attractive intermolecular forces $=>$ gas more compressible

High pressure : average distance between molecules $=>$ repulsive intermolecular forces $=>$ gas less compressible

Equation of state (3)

Compressibility factor for a real gas

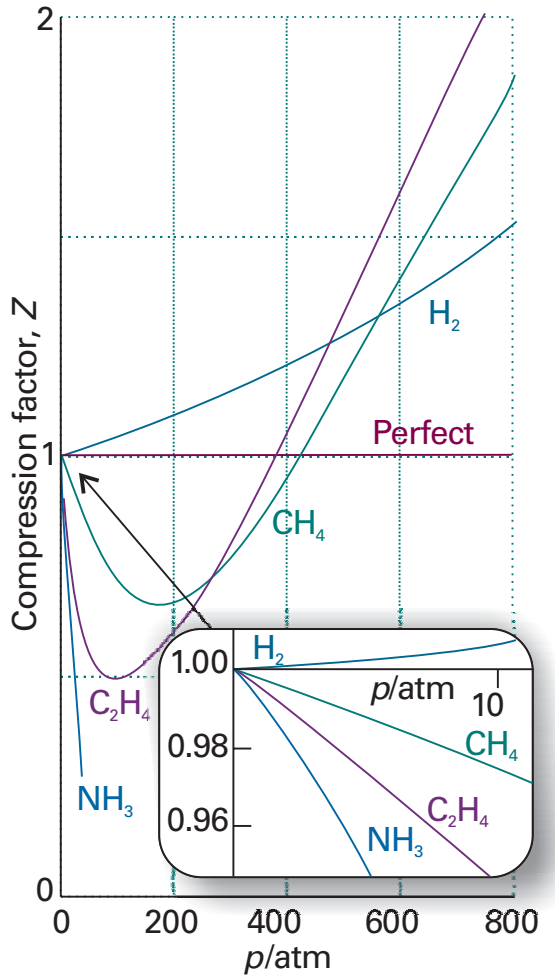
$$Z = \frac{V_m}{V_m^0} = \frac{RT}{PV_m^0}$$

Molar volume : $V_m = \frac{V}{n}$

Molar volume for perfect gas : V_m^0

Equation of state (3)

Compressibility factor for a real gas



Equation of state (4)

General form for real gas EqOS

$$P = \underbrace{\frac{RT}{V_m - b}}_{\text{Repulsion correction}} - \underbrace{\frac{a}{V_m^2}}_{\text{Attraction correction}} ;$$

a and b : account for the attractive and the repulsive intermolecular forces

a and b are gas specific constants

Equation of state (5)

Many real gas EqOS

Soave-Redlich-Kwong

$$P = \frac{RT}{V_m - b_c} - \frac{a_c \alpha}{V_m (V_m + b_c)}$$

$$a_c = 0.42747 \frac{R^2 T_c^2}{P_c}$$

$$\alpha = \left(1 + \left(0.480 + 1.574\omega - 0.176\omega^2\right)\left(1 - \sqrt{T_r}\right)\right)^2$$

$$b_c = 0.08664 \frac{RT_c}{P_c}$$

T_c, P_c **: critical temperature and pressure**

ω**=-log** $\left(P_r^{sat}\right) - 1$ **at** T_r**=0.7 : acentric factor**

P_r^{sat}**=**P^{sat}/P_c **: reduced saturation vapor pressure**

T_r**=**T/T_c **: reduced temperature.**

Equation of state (6)

Many real gas EqOS

Peng-Robinson

$$P = \frac{RT}{V_m - b_c} - \frac{a_c \alpha}{V_m(V_m + b_c) + b_c(V_m - b_c)}$$

$$a_c = 0.457235 \frac{R^2 T_c^2}{P_c}$$

$$\alpha = \left(1 + \left(0.37464 + 1.54226\omega - 0.26992\omega^2\right)\left(1 - \sqrt{T_r}\right)\right)^2$$

$$b_c = 0.077796 \frac{RT_c}{P_c}$$

Equation of state (7)

Mixing rules for Soave-Redlich-Kwong and Peng-Robinson EqOS

$$a_c = \sum_i \sum_j x_i x_j \sqrt{a_{c,i} a_{c,j}}$$

$$b_c = \sum_i x_i b_{c,j}$$

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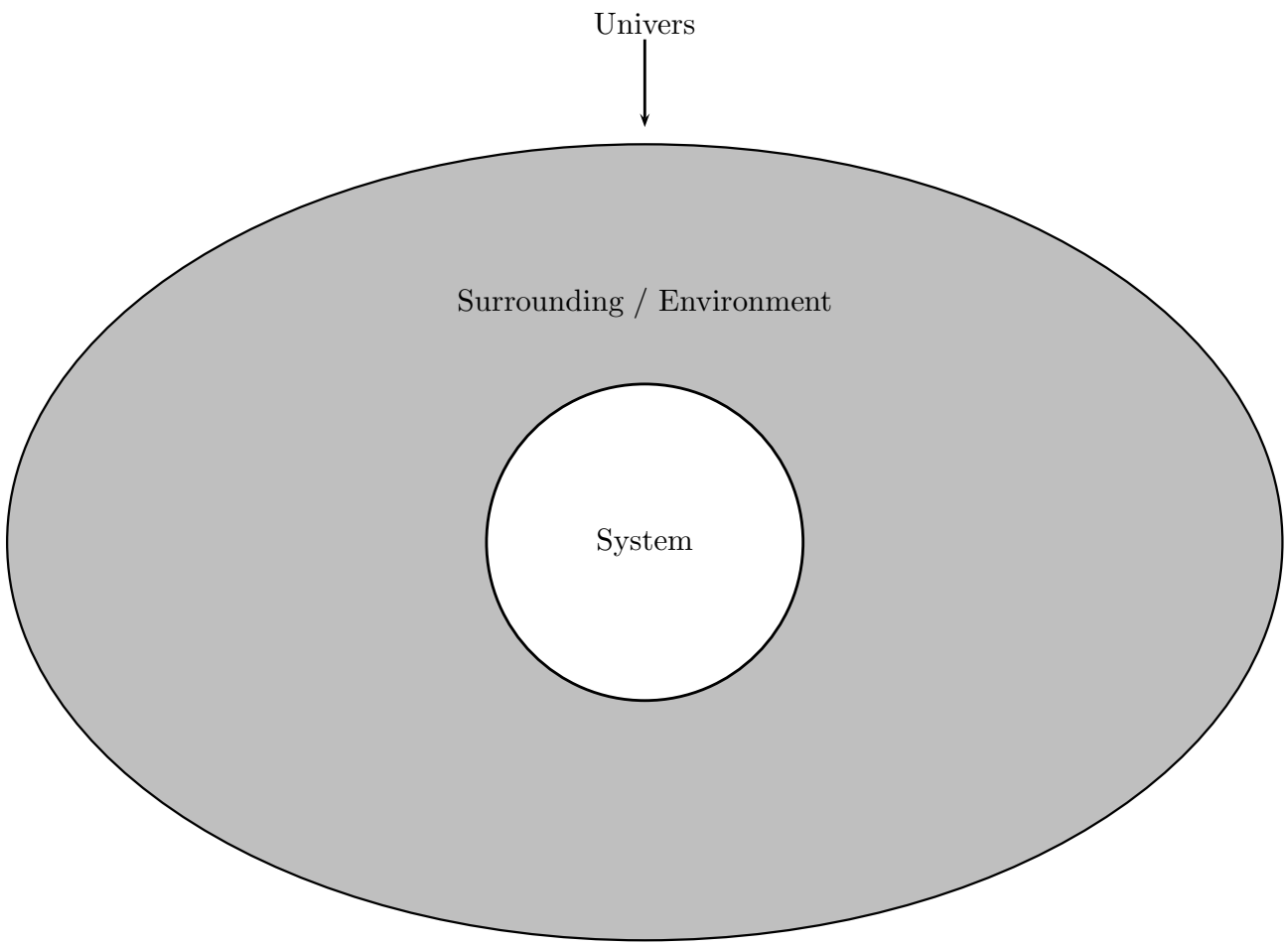
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Thermodynamic systems (1)

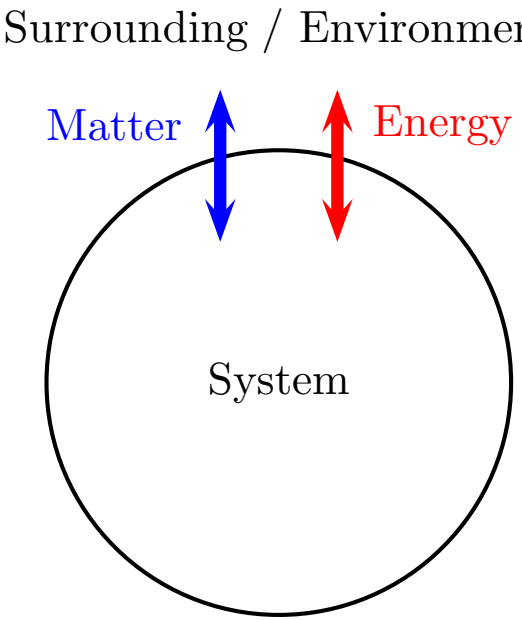
General definition



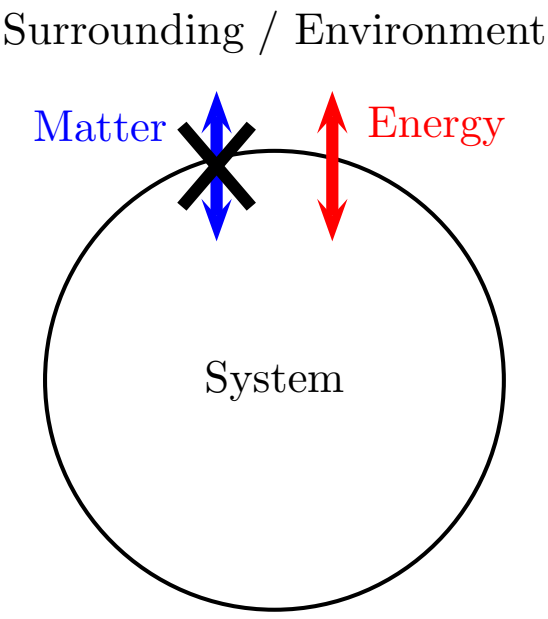
Part of the Universe where the transformation is taking place

Thermodynamic systems (2)

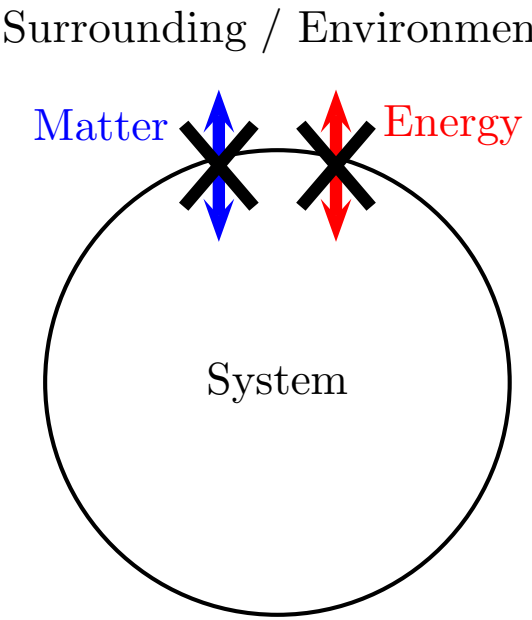
Types of system



a) Open system



b) **Closed system**

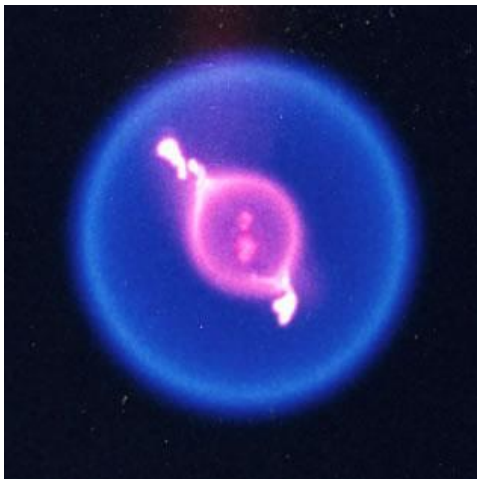


c) Isolated system

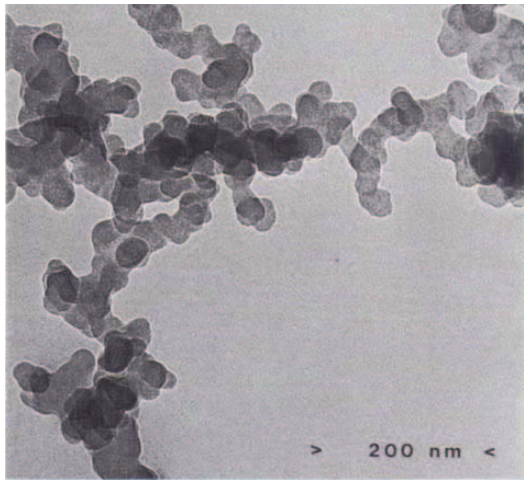
Thermodynamic systems (3)

Nature of the system

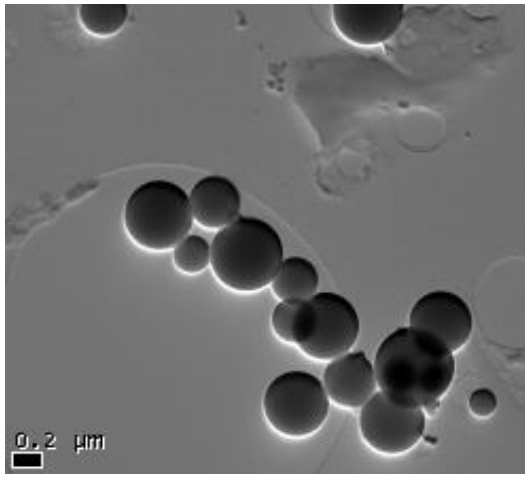
- **Homogeneous system for which there is only one phase (gas, liquid, solid)**
- **Heterogeneous system for which there are two or more phases**



a) Droplet combustion



b) **Soot formation**



c) SiO_2 formation

Thermodynamic systems (4)

Types of variables

- **Intensive variable : do not depend on the size of the system (temperature, index of refraction, the density, etc).**
- **Extensive variable : depend on the size of the system (mass, volume, etc).**

Thermodynamic systems (5)

State functions

depend on the current state of the system and are independent of the path taken between the previous and the current state

Mathematical properties

Variation between 2 states

$$\Delta F = \int_i^f F = F_f - F_i$$

Total derivative is exact

$$F_{yx} = \frac{\partial F_x}{\partial y} = \frac{\partial F}{\partial y \partial x} = \frac{\partial F}{\partial x \partial y} = \frac{\partial F_y}{\partial x} = F_{xy}$$

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Laws of Thermodynamics (1)

First law of thermodynamics

The energy of an isolated system is conserved. It can be neither created nor destroyed.

Consequences

- **Work cannot be collected without a counterpart**
- **Perpetual motion is impossible**
- **The different forms of energy are equivalent (work and heat are both energy)**

Laws of Thermodynamics (2)

Second law of thermodynamics

An isolated system evolves spontaneously from a state of non-equilibrium toward a state of equilibrium.

Consequences

- **A system evolves toward a state of highest disorder**
- **A spontaneous reaction is characterized by a creation of entropy**

Laws of Thermodynamics (3)

Third law of thermodynamics

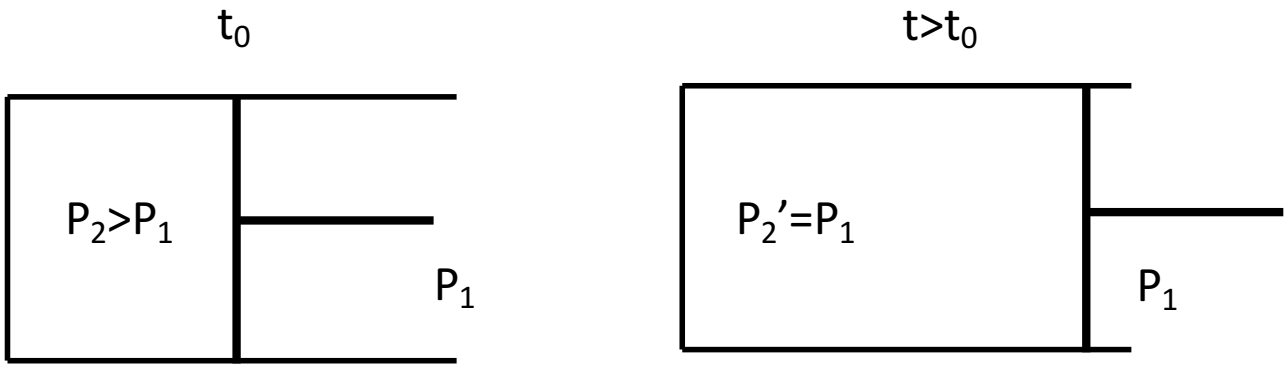
At T=0 K, all energy of thermal motion has been quenched, and in a perfect crystal all the atoms or ions are in a regular, uniform array. The entropy is zero (S=0) because there is only one way of arranging the molecule, only one micro-state is accessible : the ground state.

Important thermodynamics quantities (1)

Work (w)

Defined as motion against an opposing force.

Example : expansion of a gas



Same as in an IC engine

Energy transfer to the environment

If T_{sys} ↗, w ↗

Important thermodynamics quantities (2)

Heat (q)

Defined as a transfer of energy which originates from a difference in temperature between the system and its environment.

Two types of heat transfer processes

- **Exothermic processes : releases energy as heat into the environment**
- **Endothermic processes : acquires energy as heat from the environment**

Exothermic : $CH_4 + 2O_2 = CO_2 + 2H_2O$, $\Delta_r H^\circ = -804.29\ kJ/mol$.

Endothermic : $CH_4 = CH_3 + H$, $\Delta_r H^\circ = +438.56\ kJ/mol$.

Important thermodynamics quantities (3)

Internal energy (U)

Defined as total energy of a thermodynamic system.

Properties

- **State function**
- **Extensive property**
- **Changes :** $\Delta U = q + w$
- **Isolated system :** $\Delta U = q = w = 0$
- **Useful for constant volume processes**

Important thermodynamics quantities (4)

Internal energy : microscopic interpretation

Total energy of an isolated gas molecule (E_T)

$$E_T = \underbrace{E_{Tr} + E_R + E_V}_{\text{Thermal energy}} + \underbrace{E_{El} + E_N}_{\text{Elementary particle energy}}$$

- E_{Tr} **is the translational energy and is due to the motion of the gravity center of the molecule**
- E_R **is the rotational energy and is due to the motion of rotation of the molecule around its gravity center**
- E_V **is the vibrational energy and is due to the motion of vibration of the chemical bounds**
- E_{El} **is the electronic energy ($\simeq$ constant)**
- E_N **is the nuclear energy ($\simeq$ constant)**

Important thermodynamics quantities (5)

Mean energy of one degree of freedom of motion

$$E = \frac{1}{2} k_B T$$

with k_B : the Boltzmann constant

Translational energy (3 degrees of freedom)

$$E_{Tr} = \frac{3}{2} k_B T$$

Important thermodynamics quantities (6)

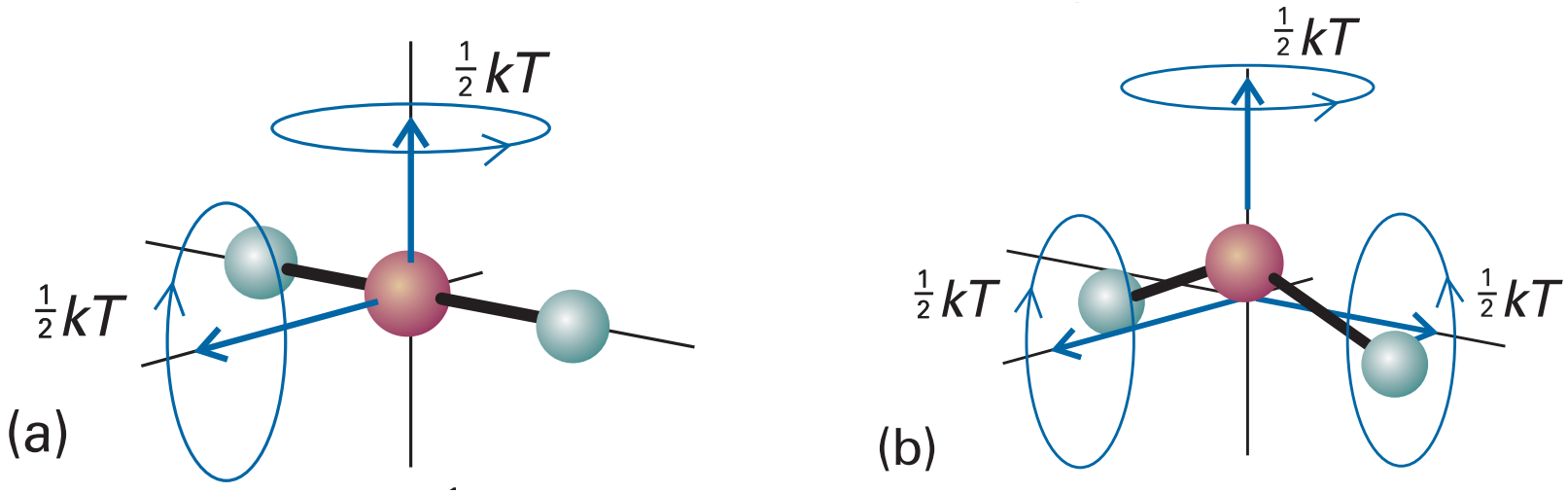
Rotational energy

For diatomic and linear polyatomic molecules

$$E_R(Di) = E_R(PolyLin) = k_B T$$

For non-linear polyatomic molecules

$$E_R(PolyNLin) = \frac{3}{2} k_B T$$



Important thermodynamics quantities (7)

Vibrational energy

For diatomic molecules

$$E_V(Di) = \underbrace{E_k}_{\text{kinetic energy}} + \underbrace{E_p}_{\text{Potential energy}} = k_B T$$

For linear polyatomic

$$E_V(PolyLin) = (3N-5)\,k_B T$$

For non-linear polyatomic

$$E_V(PolyNLin) = (3N-6)\,k_B T$$

N : number of atom in the molecule

Important thermodynamics quantities (8)

Enthalpy (H)

- **State function**
- **Extensive property**
- **Changes :** $H = U + PV$
- **Isolated system :** $\Delta H = 0$
- **Useful for constant pressure processes**

Physical interpretation

$$H + dH = U + dU + (P + dP)(V + dV) = U + dU + PV + PdV + VdP + dPdV$$

$$H + dH = H + dU + PdV + VdP$$

$$dH = dq + dw + PdV + VdP$$

$$dH = dq$$

Change of enthalpy of the system corresponds to the energy supplied as heat at constant pressure

Important thermodynamics quantities (9)

Heat capacity (C_V and C_P)

Rate of change of the internal energy

For a constant volume process

$$C_V = \left(\frac{\partial U}{\partial T}\right)_V$$

For a constant pressure process

$$C_P = \left(\frac{\partial H}{\partial T}\right)_P$$

Mayer's relationship

$$C_p - C_V = nR$$

Important thermodynamics quantities (10)

Heat capacity (C_V and C_P)

Extensive property (J.K^{-1})

$$C_V = \left(\frac{\partial U}{\partial T}\right)_V$$

Intensive property (J.mol^{-1}.K^{-1})

$$C_{V,m} = \frac{C_V}{n}$$

Specific heat capacity (J.g^{-1}.K^{-1})

$$C_{V,s} = \frac{C_V}{m}$$

Important thermodynamics quantities (11)

Physical meaning of heat capacity

$$U_m = U_m(0) + \frac{N_f}{2} k_B T$$

For a monoatomic gas

$$C_{V,m} = \frac{\partial}{\partial T}\left(U_m(0) + \frac{3}{2}RT\right) = \frac{3}{2}R = 12.47\ JK^{-1}mol^{-1}$$

For a diatomic gas

$$C_{V,m} = \frac{\partial}{\partial T}\left(U_m(0) + \frac{7}{2}RT\right) = \frac{7}{2}R = 29.1\ JK^{-1}mol^{-1}$$

Temperature change

$$\frac{dU}{C_V} = \frac{q}{C_V} = dT$$

Important thermodynamics quantities (11)

Physical meaning of heat capacity

$$U_m = U_m(0) + \frac{N_f}{2} k_B T$$

For a monoatomic gas

$$C_{V,m} = \frac{\partial}{\partial T}\left(U_m(0) + \frac{3}{2}RT\right) = \frac{3}{2}R = 12.47\ JK^{-1}mol^{-1}$$

For a diatomic gas

$$C_{V,m} = \frac{\partial}{\partial T}\left(U_m(0) + \frac{7}{2}RT\right) = \frac{7}{2}R = 29.1\ JK^{-1}mol^{-1}$$

Temperature change

$$\frac{dU}{C_V} = \frac{q}{C_V} = dT$$

=> The higher C_V, the more energy is needed to increase T

=> Part of q is used to activate internal motion (vib, rot)

Important thermodynamics quantities (12)

Infinite C_V

Phase change

$$H_2O_{Liq} = H_2O_{Vap}$$

Temperature change

$$\frac{dU}{C_V} = \frac{q}{C_V} = dT$$

During phase change, $dT = 0$

=> C_V is infinite

Important thermodynamics quantities (13)

Entropy S

$$dS = \frac{dq_{rev}}{T}$$

For spontaneous changes : $dS > 0$

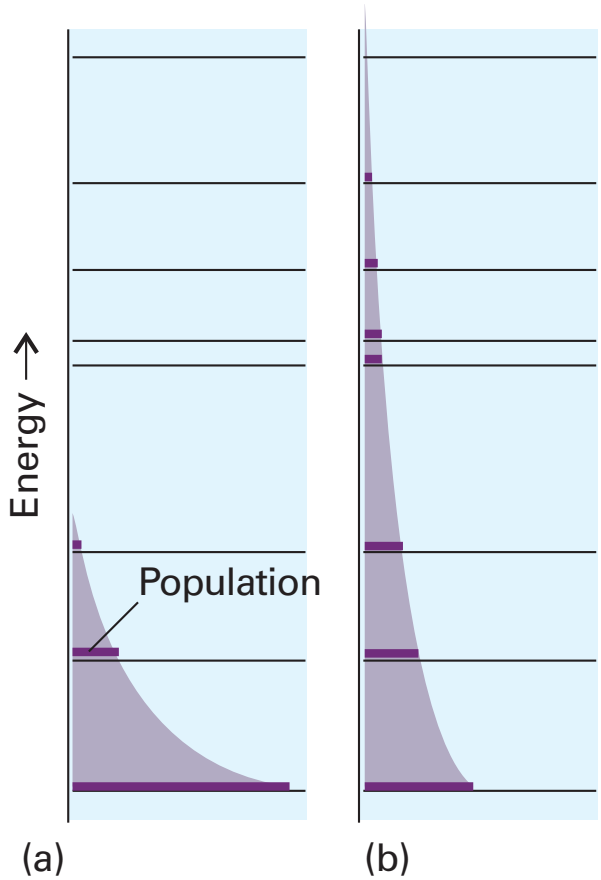
Boltzmann equation

$$S = k_B \ln W$$

As $T \nearrow$, total population of molecules or atoms is distributed over a number of energy levels (or micro-states)

Important thermodynamics quantities (14)

Population distribution



a) Low-temperature case ; b) High-temperature case

Important thermodynamics quantities (15)

Adiabatic, reversible expansion

Adiabatic change : $dq_{rev} = 0$

Work during expansion : $dw_{rev} = -PdV$

Energy changes (1)

$$dU = dq + dw = -PdV$$

and

$$dH = dU + PdV + VdP = dq + dw + PdV + VdP = VdP$$

Energy changes (2)

$$dU = nC_V dT = -PdV$$

and

$$dH = nC_P dT = VdP$$

Important thermodynamics quantities (16)

Adiabatic, reversible expansion

$$\gamma = \frac{C_P}{C_V} = -\frac{\frac{dP}{P}}{\frac{dV}{V}}$$

Integral form

$$\int \gamma \frac{dV}{V} = \int -\frac{dP}{P} + constant$$

Insentropic relationship

$$PV^{\gamma} = constant$$

With defined initial and final states

$$\int_{V_1}^{V_2} \gamma \frac{dV}{V} = \int_{P_1}^{P_2} -\frac{dP}{P}$$

Insentropic relationship

$$\frac{P_2}{P_1} = \left(\frac{V_1}{V_2}\right)^{\gamma}$$

Important thermodynamics quantities (17)

Isentropic relationships

$$\frac{T_2}{T_1} = \left(\frac{P_2}{P_1}\right)^{\frac{\gamma-1}{\gamma}} = \left(\frac{V_1}{V_2}\right)^{\gamma-1} = \left(\frac{\rho_2}{\rho_1}\right)^{\gamma-1}$$

Valid for ideal gas :

- **Perfect gas equation of gas**
- **Isentropic coefficient is constant**

Important thermodynamics quantities (18)

Helmholtz (A) and Gibbs (G) energies

$$A = U - TS$$

and

$$G = H - TS$$

Conditions for spontaneous changes :

Constant temperature and volume processes

$$dA_{T,V} \leqslant 0$$

Constant temperature and pressure processes

$$dG_{T,P} \leqslant 0$$

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Thermochemistry (1)

General definition

The study of the energy transferred as heat during the chemical reactions

Standard enthalpy changes

Notation : ΔH°

Change in enthalpy for a process in which the initial and final substances are in their standard states

The standard state of a substance at a specified temperature is its pure form at 100 kPa

Usually, 298.15 K is used as a reference temperature to report thermodynamic data.

Thermochemistry (2)

Phase transition

Water vaporization : $H_2O_{(l)} \rightarrow H_2O_{(g)}$

Notation : $\Delta_{vap}H^\circ$

Standard enthalpy = state function

Fusion : $H_2O_{(s)} \rightarrow H_2O_{(l)} \quad \Delta_{fus}H^\circ$

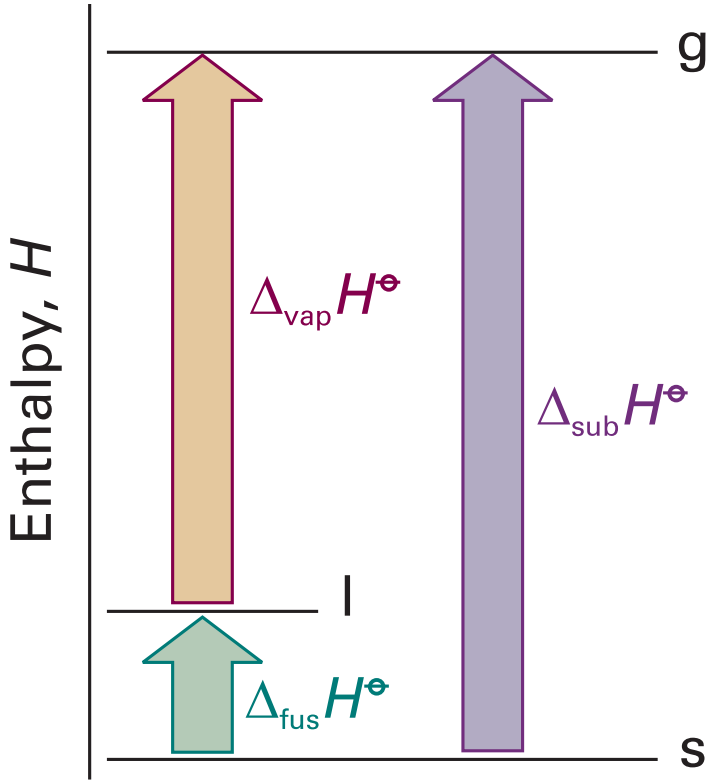
Vaporization : $H_2O_{(l)} \rightarrow H_2O_{(g)} \quad \Delta_{vap}H^\circ$

Sublimation : $H_2O_{(s)} \rightarrow H_2O_{(g)} \quad \Delta_{sub}H^\circ = \Delta_{fus}H^\circ + \Delta_{vap}H^\circ$

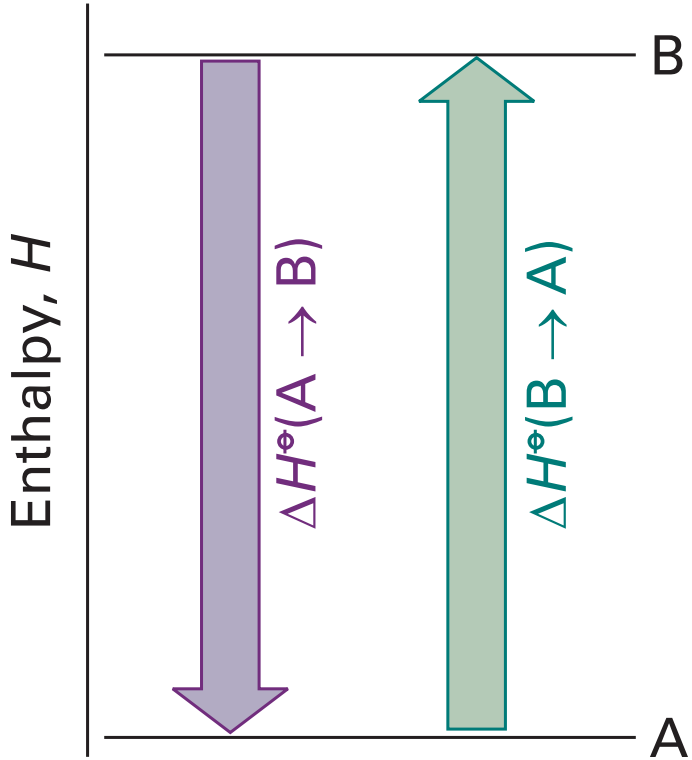
For opposite processes : $\Delta_{A\rightarrow B}H^\circ = -\Delta_{B\rightarrow A}H^\circ$

Thermochemistry (3)

Energy diagram for phase transition



a) Additivity of ΔH°



b) Opposite sign of ΔH° for opposite processes

Thermochemistry (4)

Standard enthalpy of chemical reaction

Methane combustion

$$CH_4(g) + 2O_2(g) = CO_2(g) + 2H_2O(l)$$

Standard enthalpy change

$$\Delta_r H^\circ = \Delta_f H^\circ(CO_2(g)) + 2\Delta_f H^\circ(H_2O(l)) - \Delta_f H^\circ(CH_4(g)) - 2\Delta_f H^\circ(O_2(g))$$

General form

$$\Delta_r H^\circ = \sum_{j=1}^{N_j} \nu_j \Delta_f H^\circ(P_j) - \sum_{i=1}^{N_i} \nu_i \Delta_f H^\circ(R_i)$$

Special case for formation of elements

$$\Delta_f H^\circ(H_2(g)) = \Delta_f H^\circ(O_2(g)) = \Delta_f H^\circ(N_2(g)) = 0$$

Thermochemistry (5)

Hess's law

The standard enthalpy of an overall reaction is the sum of the standard enthalpies of the individual reactions into which a reaction may be divided

Useful to obtain information about the enthalpy of a reaction which may be difficult to obtain directly

Thermochemistry (6)

Propene combustion

$$R_1 : C_3H_6(g) + \frac{9}{2}O_2(g) = 3CO_2(g) + 3H_2O(l)$$

Decomposition into several reactions

$$R_2 : C_3H_6(g) + H_2(g) = C_3H_8(g) \quad \Delta_r H^\circ(R_2) = -124\ kJ/mol$$

$$R_3 : C_3H_8(g) + 5O_2(g) = 3CO_2(g) + 4H_2O(l) \quad \Delta_r H^\circ(R_3) = -2200\ kJ/mol$$

$$R_4 : H_2O(l) = H_2(g) + \frac{1}{2}O_2(g) \quad \Delta_r H^\circ(R_4) = +286\ kJ/mol$$

Standard enthalpy of reaction

$$\Delta_r H^\circ(R_1) = \Delta_r H^\circ(R_2) + \Delta_r H^\circ(R_3) + \Delta_r H^\circ(R_4) = -124 - 2200 + 286 = -2058\ kJ/mol$$

Thermochemistry (7)

Kirchhoff's law

Describes the temperature dependence of reaction enthalpy

$$\Delta_r H^\circ(T_2) = \Delta_r H^\circ(T_1) + \int_{T_1}^{T_2} \Delta_r C_p^\circ dT$$

with

$$\Delta_r C_p^\circ = \sum_{j=1}^{N_j} \nu_j C_p^\circ(P_j) - \sum_{i=1}^{N_i} \nu_i C_p^\circ(R_i)$$

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Chemical equilibrium (1)

General considerations

Chemical reactions tend to evolve towards an equilibrium in which both reactants and products are present but have no further tendency to undergo change.

A reaction may be referred to as “complete” if the concentration of the reactants is negligibly low compared to that of the products.

In many cases, the reaction is not complete and the remaining concentration of the reactants is not negligible

Chemical equilibrium (2)

Conditions for equilibrium (1)

Chemical potential

$$\mu_i = \left(\frac{\partial G}{\partial n_i}\right)_{P,T,n_{j\neq i}}$$

For a binary mixture (A and B)

$$G = n_A\mu_A + n_B\mu_B$$

$$dG = VdP - SdT + \mu_A dn_A + \mu_B dn_B = VdP - SdT + \sum_{i=1}^{N} \mu_i dn_i$$

Consider the reaction :

$$A = B$$

$$\Delta_r G = \left(\frac{\partial G}{\partial \xi}\right)_{P,T}$$

with

ξ : the extent of reaction (in mole)

Chemical equilibrium (3)

Conditions for equilibrium (2)

At constant pressure and temperature

$$dG = \mu_A dn_A + \mu_B dn_B = -\mu_A d\xi + \mu_B d\xi = (\mu_B - \mu_A)\, d\xi$$

$$\Delta_r G = \left(\frac{\partial G}{\partial \xi}\right)_{P,T} = \mu_B - \mu_A$$

At equilibrium

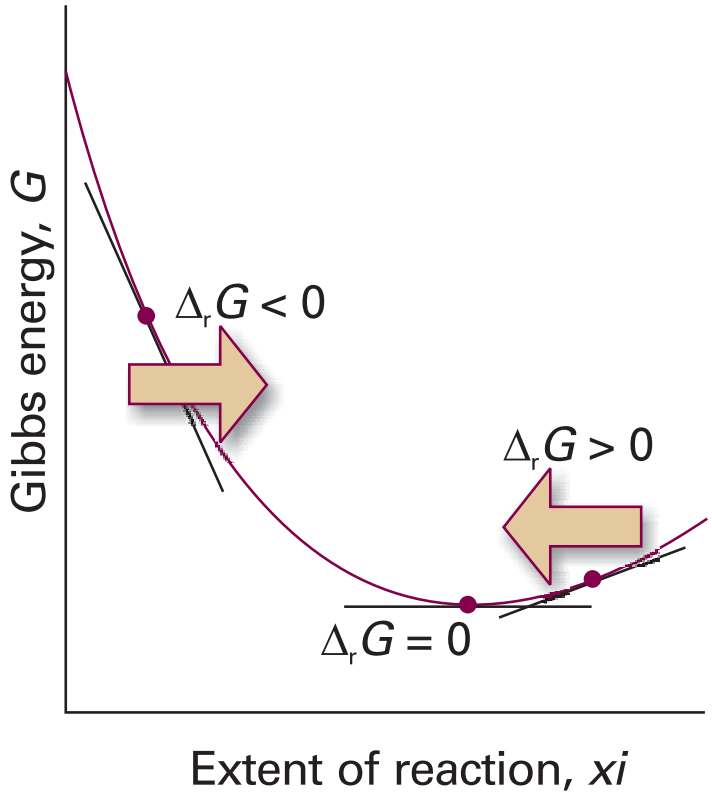
$$\Delta_r G = 0$$

$$=> \mu_A = \mu_B$$

Chemical equilibrium (4)

Conditions for equilibrium (3)

the composition at equilibrium is such that the chemical potential of the species A is equal to the chemical potential of the species B



Chemical equilibrium (5)

Description of equilibrium (1)

Consider the reaction :

$$A = B$$

Chemical potential for 1 species

$$\mu_i = \mu_i^\circ + RT \ln \frac{P_i}{P^\circ}$$

Gibbs energy

$$\Delta_r G = \left(\mu_B^\circ + RT \ln \frac{P_B}{P^\circ}\right) - \left(\mu_A^\circ + RT \ln \frac{P_A}{P^\circ}\right) = \Delta_r G^\circ + RT \ln \frac{P_B}{P^\circ} \frac{P^\circ}{P_A}$$

Reaction quotient (Q)

$$Q = \frac{P_B}{P^\circ} \frac{P^\circ}{P_A} = \frac{P_B}{P_A}$$

Chemical equilibrium (6)

Description of equilibrium (2)

At equilibrium :

$$\Delta_r G = 0$$

Equilibrium constant

$$K = \exp\left(\frac{-\Delta_r G^\circ}{RT}\right)$$

General expression for a perfect gas

$$K = \prod_{k=1}^{N_k} a_k^{\nu_k} = \exp\left(\frac{\sum_{i=1}^{N_i} \nu_i \Delta_f G^\circ(i) - \sum_{j=1}^{N_j} \nu_j \Delta_f G^\circ(j)}{RT}\right)$$

with a_k, the chemical activity

$$a_k = \frac{P_k}{P^\circ}$$

Chemical equilibrium (7)

Description of equilibrium (3)

Example :

$$H_2O(g) = H_2(g) + \frac{1}{2}O_2(g)$$

$$\Delta_r G^\circ(2300K) = 118.08\ kJ/mol$$

Advancement table

	H_2O	H_2	O_2
Initial amount	n	0	0
Equilibrium amount	$(1-\alpha)n$	αn	$\frac{1}{2}\alpha n$
Mole faction	$\frac{1-\alpha}{1+\frac{1}{2}\alpha}$	$\frac{\alpha}{1+\frac{1}{2}\alpha}$	$\frac{\frac{1}{2}\alpha}{1+\frac{1}{2}\alpha}$
Partial pressure	$P\frac{1-\alpha}{1+\frac{1}{2}\alpha}$	$P\frac{\alpha}{1+\frac{1}{2}\alpha}$	$P\frac{\frac{1}{2}\alpha}{1+\frac{1}{2}\alpha}$

Chemical equilibrium (8)

Description of equilibrium (4)

Equilibrium constant

$$K = \frac{\frac{P_{H2}}{P^\circ}\left(\frac{P_{O2}}{P^\circ}\right)^{1/2}}{\frac{P_{H2O}}{P^\circ}} = \frac{\alpha^{3/2}}{(1-\alpha)(2+\alpha)^{1/2}}\left(\frac{P}{P^\circ}\right)^{1/2}$$

Assumption : $\alpha \ll 1$

$$K = \frac{\alpha^{3/2}}{2^{1/2}}\left(\frac{P}{P^\circ}\right)^{1/2} = \exp\left(\frac{-\Delta_r G^\circ}{RT}\right)$$

We obtain

$$\alpha = \left(\left(\frac{2P^\circ}{P}\right)^{1/2}\exp\left(\frac{-\Delta_r G^\circ}{RT}\right)\right)^{2/3} = \left(2^{1/2}\exp\left(\frac{-118080}{8.314\times 2300}\right)\right)^{2/3} = 0.0205$$

Chemical equilibrium (9)

Pressure dependence of equilibrium (1)

K does not depend on pressure

$$\left(\frac{\partial K}{\partial P}\right)_T = 0$$

But composition at equilibrium can depend on pressure

Consider :

$$A = 2B$$

At equilibrium :

$n_A = (1-\alpha)n$ **and** $n_B = 2\alpha n$

Chemical equilibrium (10)

Pressure dependence of equilibrium (2)

Equilibrium constant

$$K = \frac{\alpha^2}{1-\alpha^2}\frac{P}{P^\circ}$$

Expression for α

$$\alpha = \left(\frac{1}{1+\frac{4}{K}\frac{P}{P^\circ}}\right)^{1/2}$$

α depends on pressure

$=>$ composition at equilibrium depends on pressure

Chemical equilibrium (11)

Temperature dependence of equilibrium (1)

Le Chatelier's principle

- **exothermic reactions : increased temperature favors the reactants**
- **endothermic reactions : increased temperature favors the products**

Chemical equilibrium (12)

Temperature dependence of equilibrium (2)

van't Hoff equation

$$\frac{d \ln K}{d(1/T)} = -\frac{\Delta_r H^\circ}{R}$$

Knowing K at T_1 and $\Delta_r H^\circ$, we can calculate K at T_2

$$\ln K_2 - \ln K_1 = -\frac{1}{R} \int_{1/T_1}^{1/T_2} \Delta_r H^\circ d(1/T)$$

If $\Delta_r H^\circ$ is constant in $[T_1, T_2]$

$$\ln K_2 - \ln K_1 = -\frac{\Delta_r H^\circ}{R} \left(\frac{1}{T_2} - \frac{1}{T_1} \right)$$

outlines

1 Introduction

2 State of matter and equations of state of gas

3 Thermodynamic systems

4 Thermodynamics laws and important quantities

5 Thermochemistry

6 Chemical equilibrium

7 Thermodynamics properties in combustion

Thermodynamics properties in combustion (1)

Adiabatic flame temperature (1)

Consider the reaction

$$aA + bB = cC + dD$$

Enthalpy of reaction

$$\Delta_r H = \sum_{i=1}^{N} \nu_i \left[\left(H^{\circ}_{T2} - H^{\circ}_{T0} \right) + \left(\Delta H^{\circ}_{f} \right)_{T0} \right]_i - \sum_{i=j}^{K} \nu_j \left[\left(H^{\circ}_{T1} - H^{\circ}_{T0} \right) + \left(\Delta H^{\circ}_{f} \right)_{T0} \right]_j$$

Adiabatic process

$$\Delta_r H = 0$$

Thermodynamics properties in combustion (2)

Adiabatic flame temperature (2)

Flame temperature calculated from

$$c\left[\left(H^{\circ}_{T2}-H^{\circ}_{T0}\right)\right]_C + d\left[\left(H^{\circ}_{T2}-H^{\circ}_{T0}\right)\right]_D = -\left[\left(\Delta H^{\circ}_f\right)_{T0}\right]_C - \left[\left(\Delta H^{\circ}_f\right)_{T0}\right]_D + a\left[\left(H^{\circ}_{T1}-H^{\circ}_{T0}\right)+\left(\Delta H^{\circ}_f\right)_{T0}\right]_A + b\left[\left(H^{\circ}_{T1}-H^{\circ}_{T0}\right)+\left(\Delta H^{\circ}_f\right)_{T0}\right]_B$$

with tabulated values of

$$\left(H^{\circ}_{T}-H^{\circ}_{T0}\right)$$

Thermodynamics properties in combustion (3)

Adiabatic flame temperature (3)

Example with T_1=200 K

$$H_2 + \frac{1}{2}O_2 = H_2O$$

Enthalpy of reaction

$$\Delta_r H = \left[\left(H^\circ_{T2} - H^\circ_{T0}\right) + \left(\Delta H^\circ_f\right)_{T0}\right]_{H2O} - \left[\left(H^\circ_{T1} - H^\circ_{T0}\right) + \left(\Delta H^\circ_f\right)_{T0}\right]_{H2} - \frac{1}{2}\left[\left(H^\circ_{T1} - H^\circ_{T0}\right) + \left(\Delta H^\circ_f\right)_{T0}\right]_{O2}$$

By definition

$$\left(\Delta H^\circ_f\left(H_2\right)\right)_{T0} = \left(\Delta H^\circ_f\left(O_2\right)\right)_{T0} = 0$$

and

$$\Delta_r H = 0$$

Thermodynamics properties in combustion (4)

Adiabatic flame temperature (4)

Final expression

$$\left[\left(H^{\circ}_{T2} - H^{\circ}_{T0}\right)\right]_{H2O} = -\left[\left(\Delta H^{\circ}_{f}\right)_{T0}\right]_{H2O} + \left[\left(H^{\circ}_{T1} - H^{\circ}_{T0}\right)\right]_{H2} + \frac{1}{2}\left[\left(H^{\circ}_{T1} - H^{\circ}_{T0}\right)\right]_{O2}$$

From thermodynamic database, we know that

- $\left[\left(\Delta H^{\circ}_{f}\right)_{T0}\right]_{H2O}$=-243.83 kJ/mol
- $\left[\left(H^{\circ}_{T1} - H^{\circ}_{T0}\right)\right]_{H2}$=-2.77 kJ/mol
- $\frac{1}{2}\left[\left(H^{\circ}_{T1} - H^{\circ}_{T0}\right)\right]_{O2}$=-2.87 kJ/mol

Thus

$$\left[\left(H^{\circ}_{T2} - H^{\circ}_{T0}\right)\right]_{H2O} = 237.6 \quad \text{kJ/mol.}$$

Thermodynamics properties in combustion (5)

Adiabatic flame temperature (5)

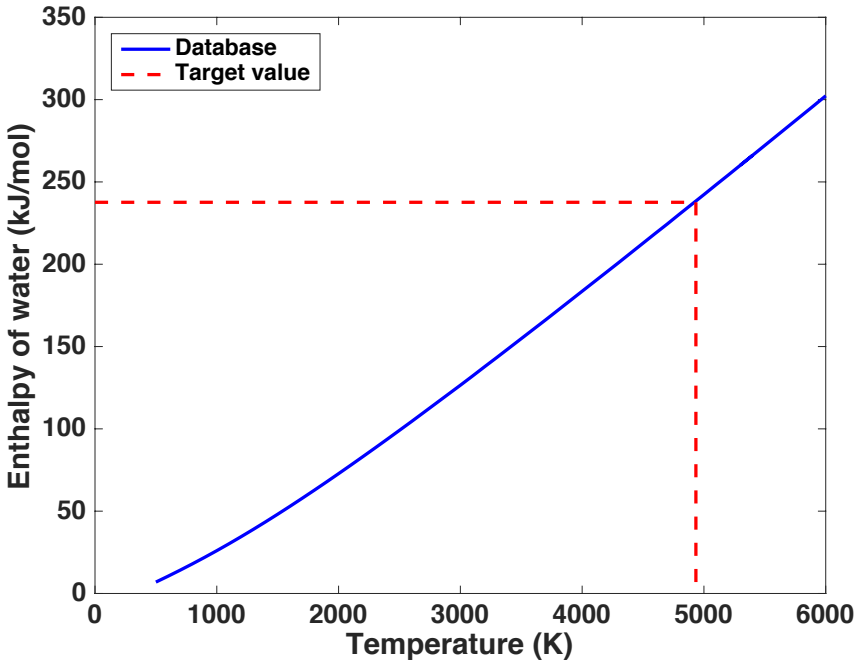
Thermodynamic table

Temperature (K)	H-H_{298} (kJ/mol)
4000	183.6
4100	189.4
4200	195.2
4300	201
4400	206.9
4500	212.8
4600	218.7
4700	224.5
4800	230.5
4900	236.4
5000	242.3

Thermodynamics properties in combustion (6)

Adiabatic flame temperature (6)

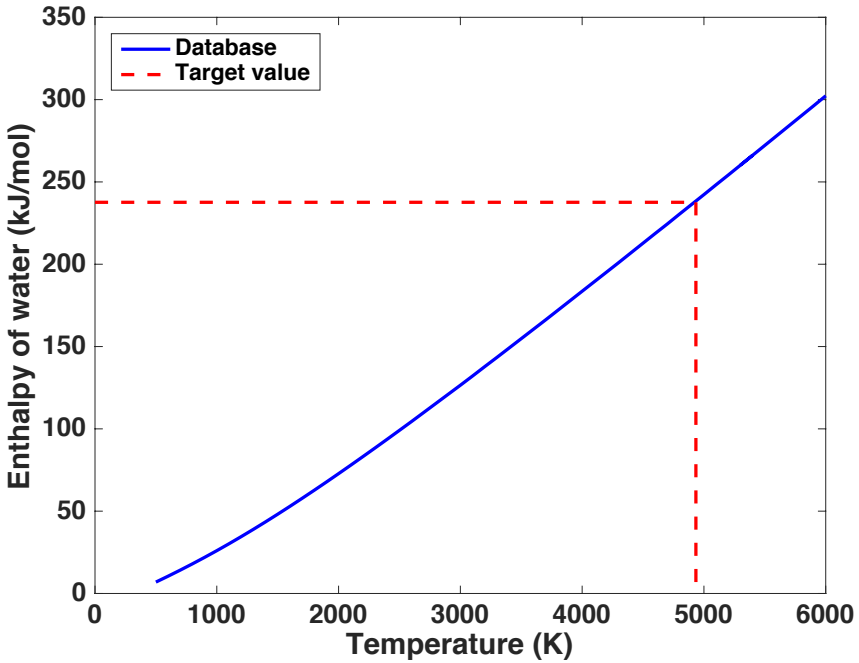
Graphical determination



Thermodynamics properties in combustion (6)

Adiabatic flame temperature (6)

Graphical determination

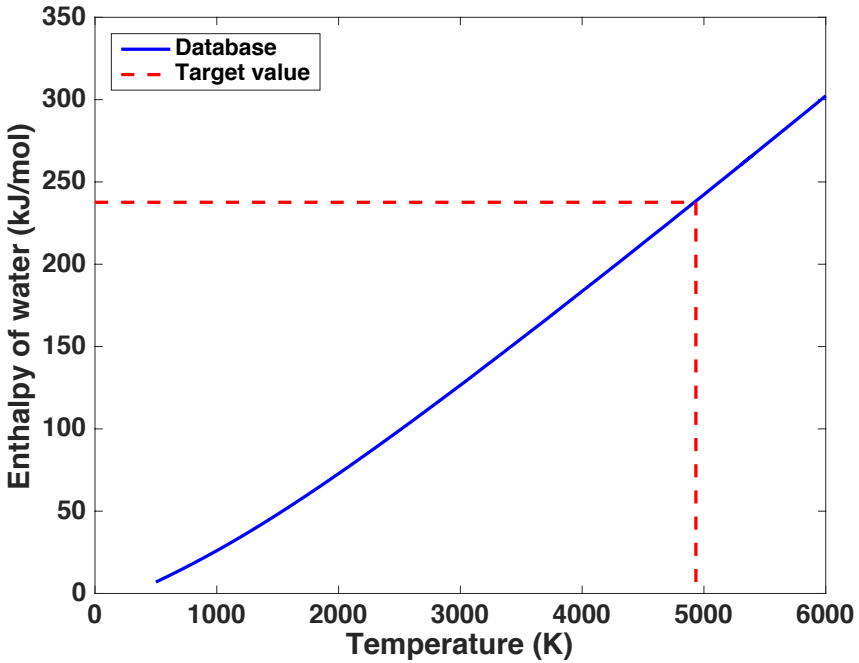


=> T_f=4934 K

Thermodynamics properties in combustion (6)

Adiabatic flame temperature (6)

Graphical determination



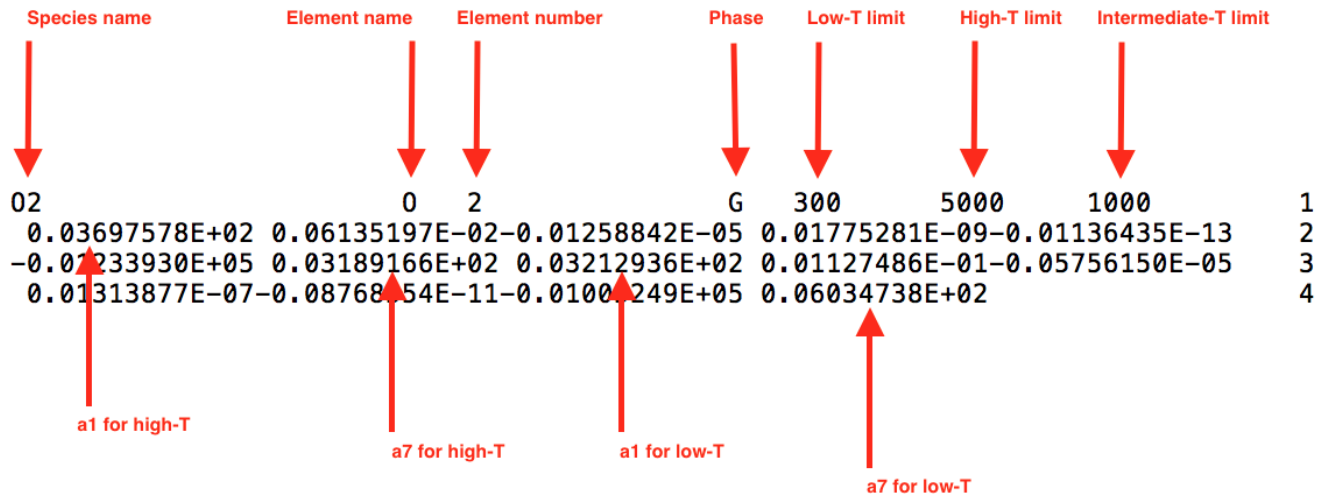
=> T_f=4934 K

In reality, $T_f \simeq$3100 K

Thermodynamics properties in combustion (7)

Declaring thermodynamic data

In combustion software, polynomial description



- **Heat capacity :** $\frac{C_P}{R} = a_1 + a_2 T + a_3 T^2 + a_4 T^3 + a_5 T^4$
- **Enthalpy :** $\frac{H^\circ}{RT} = a_1 + \frac{a_2}{2} T + \frac{a_3}{3} T^2 + \frac{a_4}{4} T^3 + \frac{a_5}{5} T^4 + \frac{a_6}{T}$
- **Entropy :** $\frac{S^\circ}{R} = a_1 \ln T + a_2 T + \frac{a_3}{2} T^2 + \frac{a_4}{3} T^3 + \frac{a_5}{4} T^4 + a_7$

Thermodynamics properties in combustion (8)

Chemkin : a combustion software (1)

chem.inp - Notepad

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```
ELEM  O N AR H
END
SPECIES ! structure, source of thermo-data, source of transport
H  ! burcat, chemkin
H2 ! burcat, chemkin
O   O2    H2O    OH    H2O2    HO2
AR ! burcat, chemkin
N2 ! burcat, chemkin

END
REACTIONS

! H2 REACTIONS

H2+M=H+H+M                4.57E+19  -1.40     104380       !1          Tsang &
   H2/2.5/ H2O/12/ AR/0.0/
H2+AR=H+H+AR              5.84E+18  -1.10     104380       !2          Tsang &
H2+O2=OH+OH               2.50E+12   0.00      39000       !3          Konnov
H+H+H=H2+H                3.20E+15   0.00          0       !4          Konnov
H+HO2=H2O+O               3.00E+13   0.00       1720       !5          Konnov


! O REACTIONS

O+H2=H+OH                 5.08E+04   2.67       6290       !6          Sutherl
O+O+M=O2+M                6.16E+15  -0.50          0       !7          Tsang &
   H2/2.5/ H2O/12/ AR/0.0/
O+O+AR=O2+AR              1.89E+13   0.00      -1788       !8          Tsang &


! H REACTIONS

H+O2=O+OH                 1.91E+14   0.00      16439       !9          Pirragl
H+O2(+M)=HO2(+M)          1.48E+12   0.60          0       !10         Mueller
LOW/3.482E+16  -0.411  -1115/
TROE/0.5  1E-30  1E30/
H2/2.5/ H2O/12/ AR/0.0/
H+O2(+AR)=HO2(+AR)        1.48E+13   0.60          0       !11         TJK bas
LOW/1.49E+15   0.00  -1000/
TROE/0.45   1E-30  1E+30/
H+O+M=OH+M                4.71E+18  -1.00          0       !12         Tsang &
H2/2.5/ H2O/12/ AR/.75/


! OH REACTIONS

OH+H2=H2O+H               2.16E+08   1.51       3430       !13         Michael
H2O+O=OH+OH               2.97E+06   2.02      13400       !14         Sutherl
H2O2(+M)=OH+OH(+M)        2.95E+14   0.00      48430       !15         Brouwer
LOW/1.20E+17 0.00 45500/
```

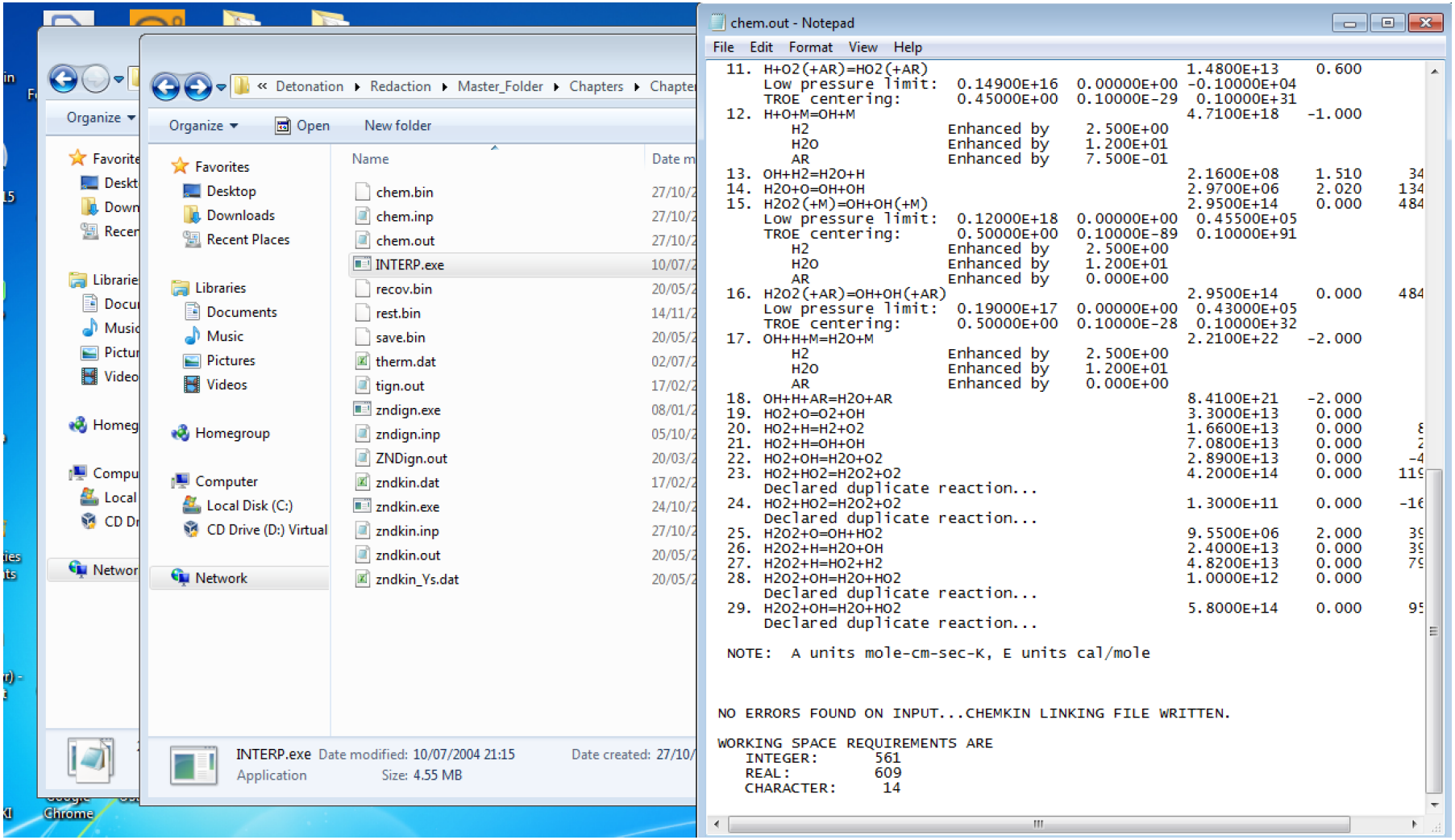
therm.dat - Notepad

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```
THERMO ALL
 300.000  1000.000  5000.000
H                 L 6/94H   10    00    00    0G    200.000  6000.00  1000.0
 0.25000000E+01 0.00000000E+00 0.00000000E+00 0.00000000E+00 0.00000000E+00
 0.25473660E+05-0.44668285E+00 0.25000000E+01 0.00000000E+00 0.00000000E+00
 0.00000000E+00 0.00000000E+00 0.25473660E+05-0.44668285E+00 0.26219035E+05
H2  REF ELEMENT   RUS 78H   20    00    00    0G    200.000  6000.00  1000.0
 0.29328305E+01 0.82659802E-03-0.14640057E-06 0.15409851E-10-0.68879615E-15
-0.81305582E+03-0.10243164E+01 0.23443029E+01 0.79804248E-02-0.19477917E-04
 0.20156967E-07-0.73760289E-11-0.91792413E+03 0.68300218E+00 0.00000000E+00
O                 L 1/90O   10    00    00    0G    200.000  6000.00  1000.0
 2.54363697E+00-2.73162486E-05-4.19029520E-09 4.95481845E-12-4.79553694E-16
 2.92260120E+04 4.92229457E+00 3.16826710E+00-3.27931884E-03 6.64306396E-06
-6.12806624E-09 2.11265971E-12 2.91222592E+04 2.05193346E+00 2.99687009E+04
O2 REF ELEMENT    RUS 89O   20    00    00    0G    200.000  6000.00  1000.0
 3.66096083E+00 6.56365523E-04-1.41149485E-07 2.05797658E-11-1.29913248E-15
-1.21597725E+03 3.41536184E+00 3.78245636E+00-2.99673415E-03 9.84730200E-06
-9.68129508E-09 3.24372836E-12-1.06394356E+03 3.65767573E+00 0.00000000E+00
OH                O   1H    10    00    0G    200.000  6000.00  1000.0
 2.83853033E+00 1.10741289E-03-2.94000209E-07 4.20698729E-11-2.42289890E-15
 3.69780808E+03 5.84494652E+00 3.99198424E+00-2.40106655E-03 4.61664033E-06
-3.87916306E-09 1.36319502E-12 3.36889836E+03-1.03998477E-01 4.48615380E+03
HO2               L 5/89H   1O    20    00    0G    200.000  6000.00  1000.0
 0.41722659E+01 0.18812098E-02-0.34629297E-06 0.19468516E-10 0.17609153E-15
 0.61818851E+02 0.29577974E+01 0.43017880E+01-0.47490201E-02 0.21157953E-04
-0.24275961E-07 0.92920670E-11 0.29480876E+03 0.37167010E+01 0.15096500E+04
H2O               L 5/89H   2O    10    00    0G    200.000  6000.00  1000.0
 0.26770389E+01 0.29731816E-02-0.77376889E-06 0.94433514E-10-0.42689991E-14
-0.29885894E+05 0.68825500E+01 0.41986352E+01-0.20364017E-02 0.65203416E-05
-0.54879269E-08 0.17719680E-11-0.30293726E+05-0.84900901E+00-0.29084817E+05
H2O2              L 2/93H   2O    20    00    0G    200.000  6000.00  1000.0
 4.57977305E+00 4.05326003E-03-1.29844730E-06 1.98211400E-10-1.13968792E-14
-1.80071775E+04 6.64970694E-01 4.31515149E+00-8.47390622E-04 1.76404323E-05
-2.26762944E-08 9.08950158E-12-1.77067437E+04 3.27373319E+00-1.63425145E+04
AR REF ELEMENT    L 6/88AR  1    0     0     0G    200.000  6000.00  1000.0
 0.25000000E+01 0.00000000E+00 0.00000000E+00 0.00000000E+00 0.00000000E+00
-0.74537500E+03 0.43796749E+01 0.25000000E+01 0.00000000E+00 0.00000000E+00
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```

Thermodynamics properties in combustion (9)

Chemkin : a combustion software (2)



Thermodynamics properties in combustion (10)

Chemkin application : adiabatic flame temperature

Consider the reaction

$$H_2 + \frac{1}{2}O_2 = H_2O$$

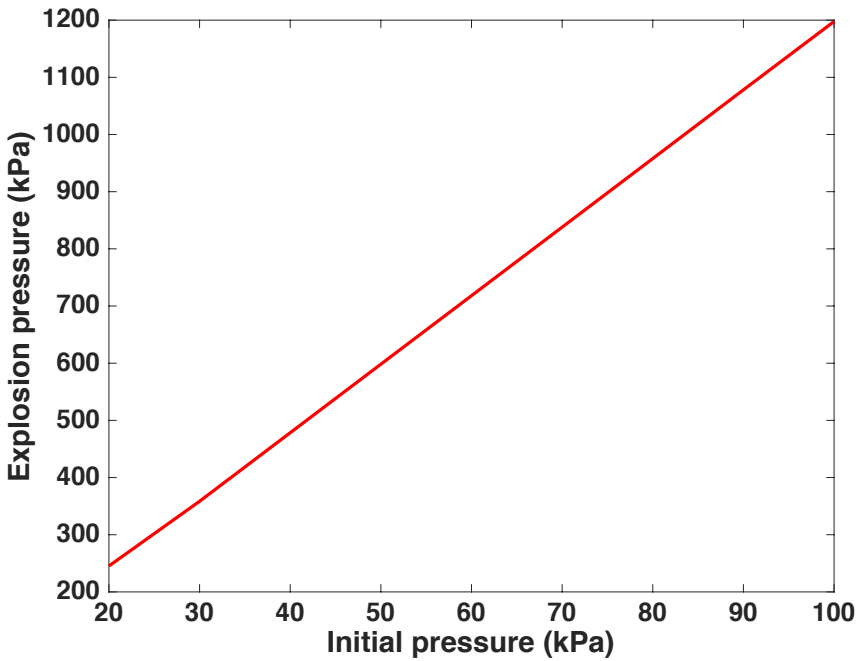
Species included	Flame temperature (K)
H_2, O_2, H_2O	3510.9
H_2, O_2, H_2O H, O, OH	3079.0
H_2, O_2, H_2O H, O, OH HO_2, H_2O_2	3078.9

Thermodynamics properties in combustion (11)

Chemkin application : experiment design

Consider the mixture

$H_2 + N_2O$ at P_1=20-100 kPa and T_1=300 K

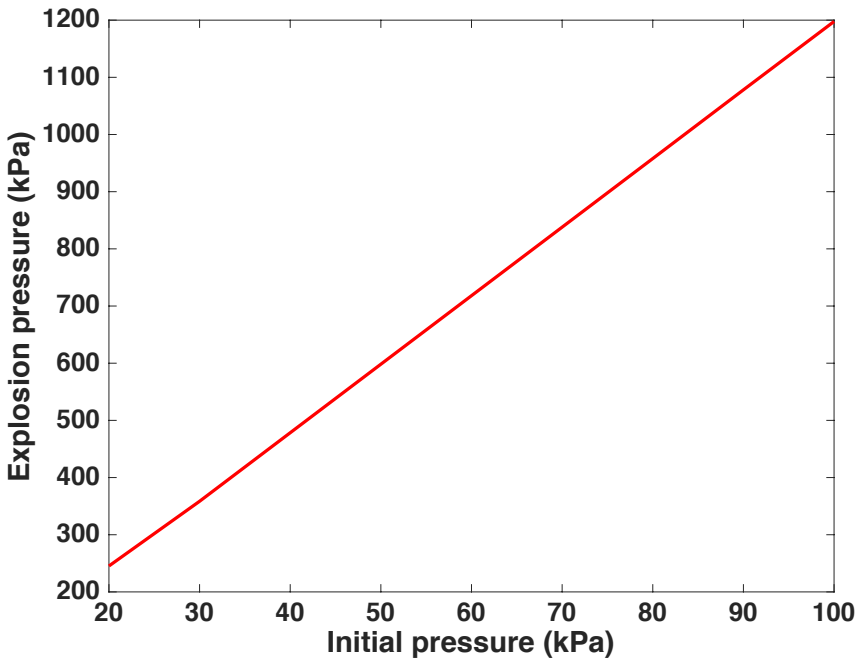


Thermodynamics properties in combustion (11)

Chemkin application : experiment design

Consider the mixture

$H_2 + N_2O$ at P_1=20-100 kPa and T_1=300 K



=> combustion vessel should be designed to resist a pressure much above P_{max}=1.2 MPa.

Thermodynamics properties in combustion (12)

Chemkin application : experiment with toxic mixture (1)

Flame retardant : CF_3I

Formation of HF : lethal concentration $\simeq$ 1000 ppm

Combustion of C_4H_{10}-Air-CF_3I mixture

at T_1=300 K and P_1=101 kPa

Thermodynamics properties in combustion (13)

Chemkin application : experiment with toxic mixture (2)

Equilibrium composition

MIXTURE:	INITIAL STATE:	EQUILIBRIUM STATE:
P (atm)	1.0000E+00	1.0000E+00
T (K)	3.0000E+02	1.6299E+03
V (cm3/gm)	7.7787E+02	5.5034E+03
H (erg/gm)	-3.5216E+09	-3.5216E+09
U (erg/gm)	-4.3098E+09	-9.0980E+09
S (erg/gm-K)	6.5881E+07	1.0271E+08
W (gm/mol)	3.1647E+01	2.4300E+01
Mol Fractions		
H	0.0000E+00	2.8015E-05
H2	0.0000E+00	1.4293E-01
O	0.0000E+00	1.1075E-10
O2	1.9733E-01	2.8974E-11
OH	0.0000E+00	7.9976E-07
H2O	0.0000E+00	8.2205E-02
N2	7.3588E-01	5.6504E-01
CO	0.0000E+00	1.6148E-01
CO2	0.0000E+00	2.9677E-02
CH2O	0.0000E+00	1.0379E-08
HCO	0.0000E+00	4.6035E-09
HOCHO	0.0000E+00	1.8994E-09
OCHO	0.0000E+00	1.0163E-12
CH4	0.0000E+00	6.0266E-09
CH3	0.0000E+00	2.4309E-11
C4H10	6.0720E-02	0.0000E+00
HF	0.0000E+00	1.3982E-02
F	0.0000E+00	1.8295E-10
CHF:O	0.0000E+00	2.4694E-10
CF:O	0.0000E+00	1.1060E-12
I2	0.0000E+00	1.7962E-06
HI	0.0000E+00	1.9477E-03
I	0.0000E+00	2.7095E-03
IO	0.0000E+00	2.0206E-11
IOH	0.0000E+00	1.6240E-08
CF3I	6.0700E-03	1.8510E-25

Thermodynamics properties in combustion (13)

Chemkin application : experiment with toxic mixture (2)

Equilibrium composition

MIXTURE:	INITIAL STATE:	EQUILIBRIUM STATE:
P (atm)	1.0000E+00	1.0000E+00
T (K)	3.0000E+02	1.6299E+03
V (cm3/gm)	7.7787E+02	5.5034E+03
H (erg/gm)	-3.5216E+09	-3.5216E+09
U (erg/gm)	-4.3098E+09	-9.0980E+09
S (erg/gm-K)	6.5881E+07	1.0271E+08
W (gm/mol)	3.1647E+01	2.4300E+01
Mol Fractions		
H	0.0000E+00	2.8015E-05
H2	0.0000E+00	1.4293E-01
O	0.0000E+00	1.1075E-10
O2	1.9733E-01	2.8974E-11
OH	0.0000E+00	7.9976E-07
H2O	0.0000E+00	8.2205E-02
N2	7.3588E-01	5.6504E-01
CO	0.0000E+00	1.6148E-01
CO2	0.0000E+00	2.9677E-02
CH2O	0.0000E+00	1.0379E-08
HCO	0.0000E+00	4.6035E-09
HOCHO	0.0000E+00	1.8994E-09
OCHO	0.0000E+00	1.0163E-12
CH4	0.0000E+00	6.0266E-09
CH3	0.0000E+00	2.4309E-11
C4H10	6.0720E-02	0.0000E+00
HF	0.0000E+00	1.3982E-02
F	0.0000E+00	1.8295E-10
CHF:O	0.0000E+00	2.4694E-10
CF:O	0.0000E+00	1.1060E-12
I2	0.0000E+00	1.7962E-06
HI	0.0000E+00	1.9477E-03
I	0.0000E+00	2.7095E-03
IO	0.0000E+00	2.0206E-11
IOH	0.0000E+00	1.6240E-08
CF3I	6.0700E-03	1.8510E-25

X_{HF}=14%, 50 to 70 times the lethal concentration

Thermodynamics properties in combustion (14)

Chemkin application : experiment with toxic mixture (3)

Equilibrium composition

MIXTURE:	INITIAL STATE:	EQUILIBRIUM STATE:
P (atm)	1.0000E+00	1.0000E+00
T (K)	3.0000E+02	1.6299E+03
V (cm3/gm)	7.7787E+02	5.5034E+03
H (erg/gm)	-3.5216E+09	-3.5216E+09
U (erg/gm)	-4.3098E+09	-9.0980E+09
S (erg/gm-K)	6.5881E+07	1.0271E+08
W (gm/mol)	3.1647E+01	2.4300E+01
Mol Fractions		
H	0.0000E+00	2.8015E-05
H2	0.0000E+00	1.4293E-01
O	0.0000E+00	1.1075E-10
O2	1.9733E-01	2.8974E-11
OH	0.0000E+00	7.9976E-07
H2O	0.0000E+00	8.2205E-02
N2	7.3588E-01	5.6504E-01
CO	0.0000E+00	1.6148E-01
CO2	0.0000E+00	2.9677E-02
CH2O	0.0000E+00	1.0379E-08
HCO	0.0000E+00	4.6035E-09
HOCHO	0.0000E+00	1.8994E-09
OCHO	0.0000E+00	1.0163E-12
CH4	0.0000E+00	6.0266E-09
CH3	0.0000E+00	2.4309E-11
C4H10	6.0720E-02	0.0000E+00
HF	0.0000E+00	1.3982E-02
F	0.0000E+00	1.8295E-10
CHF:O	0.0000E+00	2.4694E-10
CF:O	0.0000E+00	1.1060E-12
I2	0.0000E+00	1.7962E-06
HI	0.0000E+00	1.9477E-03
I	0.0000E+00	2.7095E-03
IO	0.0000E+00	2.0206E-11
IOH	0.0000E+00	1.6240E-08
CF3I	6.0700E-03	1.8510E-25

Special equipped needed

or

Don’t do the experiments